



Gel to regenerate cornea

IIT-Hyderabad research team develops hydrogel to regenerate cornea.
<https://digit.in/oct21-69>



Breakthrough in battery recycling

Breakthrough research makes battery recycling more economical.
<https://digit.in/oct21-70>



QUANTUM MATERIALS

Playing with science to cut down power bills.

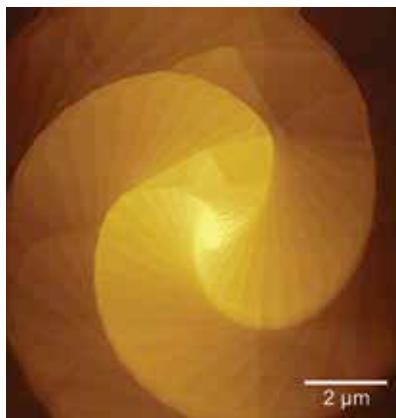
Satvik Pandey | feedback@digit.in

All the scientific and technological developments that we have seen occur around us through history have gone through three main phases. The first one is where the core concept development and prototype testing happens, the second, where the concept moves out of the labs and onto the mass production lines as a commodity of public use, and the third, where researchers make attempts to make things smaller.

In the larger scheme of things, Quantum materials are a part of that stage of technological development of making the technology smaller and smaller. Quantum materials can help us improve the quality of life in ways that we are yet to think of.

MAKE 'EM TWIST!

A team of scientists, led by Professor Song Jin of the University of Wisconsin, were successful in developing a way using which they could control the growth and twisting of quantum materials. The way that quantum materials were twisted before this discovery was by stacking two sheets of quantum material over each other and then controlling the degree of twisting by using hands. The main advantage of twisting quantum



These spirals have superconductive properties

materials is that at a certain degree of twisting, which is usually only a few degrees, quantum materials, tend to lose and gain certain properties. For example, at a certain degree of twisting, when put in low temperatures, quantum materials tend to lose their electrical resistance. This means that electricity can flow through them, without any loss and hindrance, at close to 100 per cent efficiency. As told by Professor Jin, in his statement regarding this breakthrough discovery, "Researchers consider these 2D-quantum materials, and call such work 'twistronics.'"

During the process of discovering this new phenomenon, the main challenge that lay in the way of Professor Lin and his team was of uncontrolled growth of 2D quantum materials. They could not control the angle of twisting between the layers of 2D quantum materials, which were being made to grow by themselves. This was because the interaction between the layers was found to be very weak. Here, thinking outside the box helped the team of researchers when they decided to shift their focus away from Euclidian geometry.

They were finally able to develop a non-Euclidean spiral structure and make the spirals twist in a way which they would have a certain degree of control over and would also be able to predict its structure. These twisting spirals exhibited properties that had never been seen before and would help in studying quantum particles at a nanoscale.

HIGH VOLTAGE ACTION

In another development in the electrical and technological deployment of quantum materials, the researchers from the University of Buffalo, New York, have reportedly been successful in making a two-dimensional transistor made from graphene and molybdenum disulphide (MoS₂). The distinguishing characteristic of this newly made cutting edge technology is that this transistor can meet the power demand of power-hungry